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Computer Vision

1) Computer Vision enables Computers to capture and analyse, it gives the outcome of ~~Visual Image~~.

(a) Main objective:

→ To preprocess the Image for noise ~~cancellation~~ and enhancements.

→ To apply the feature extraction to the people.

→ Then, predict the outcome using Computer Vision.

(b) Two-Real world Application:

→ Attendance Marking, System.

→ plant disease analysis. as d

2) The key features of Computer ~~Vision~~ that makes a difference in Image, are by accurate and analyze or capture the Image Pixel, block-by-block.

(a) Feature Extraction:

Feature Extraction That applies filters (Such as brightness, Saturation, Contrast) to the pixel or values to the Image.

(b) object Recognition:

By Using Such as feature extraction like enhancing pixels, Intensity, Contrast to the pixel, it can Recognize the object without any issues. So, this helps to Improve Computer Vision System.

3) Image Processing:

The different levels of Image processing are Low-level, Mid-level and High-level are used in the Computer Vision.

(a) Low-level Image Processing:

It applies Image processing to the Pixels or by changing Intensity value, it can enhance the Image levels to the System needs.

b) Different b/w Mid-level and High-level:

- * Image Segmentation and feature extraction.
- * object Recognition and Scene Understanding.

Digital Image:

Digital Image is represented as a matrix of Pixel arranged in Rows and Columns.
as Pixel:

Pixel is the Smallest Unit of a digital image that stores Color or Intensity.

b) Image Resolution:

High resolution Contains more pixels, resulting in better quality of image.

Image Sensing and acquisition:

Image Sensing Captures light from a scene, and Image acquisition Converts it into a digital image.

a) Image Sensor:

It Converts Incoming light into electrical signals, which converted into a digital image.

b) Two Image acquisition devices:

* Digital ~~Scanner~~ Camera.

* Scanners

6) Sampling and Quantization:

Sampling is the no. of pixels, while quantization determines the intensity values assigned to pixels.

(a) Sampling

is the process of converting a Continuous Image into discrete Pixels.

(b) Quantization in Image processing:

Quantization assigns Intensity or Color to the values to each Sampled Pixels.

7) Image Formation:

Image Formation it explains how light from an object passes through a Camera and form an Image.

(a) perspective projection:

The process of projecting a 3D scene into a 2D Image onto image plane while preserving depth.

(b) Role of illumination:

Illumination provides light required for capturing Images and affect brightness, and image quality.

Image Processing

It enhances or modifies Images, while
Computer Vision Interprets and Understands Image
Content.

a) Image Processing:

The manipulation and enhancement of
digital image using Computational techniques.

b) Image Processing and CV

IP:

- * Enhance or modifies Image.
- * Output is usually another Image.

CV:

- * Understands and interprets Image.
- * Output is Information or decision.

Q) Computer Vision automates Image analysis in
healthcare, transportation, Security and
manufacturing.

a) Application in healthcare:

CV analyzes medical Images Such as
X-rays, CT scans and MRI Scans to detect
diseases early.

b) Autonomous Vehicles:

It detects lanes, traffic signal, pedestrians and other vehicles to enable safe navigation.

10) Digital Image provide the basic representation and quality required for accurate image analysis.

a) Pixel Stores:

The Intensity and color information that together form the complete digital image.

b) Sampling: determines the no. of Pixels captured.

* Image Resolution: determines the level of detail, while higher resolution improving detection and accuracy.